

YUHAN JIANG

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EDUCATION

EIT Manufacturing Master & Doctoral School Data Science and AI for a Competitive Manufacturing Master	Sep 2023 - Mar 2026
Università di Trento Master of Science - MSc, Computer Science Master	Sep 2023 - Mar 2026
University of Applied Sciences and Arts of Southern Switzerland Master of Science - MSc, Engineering Master	Sep 2023 - Mar 2026
Anhui University (Project 211) Intelligent Science and Technology • GPA: 3.16/5.00 (Average Score: 84.34/100) • Awards: Second Prize for Outstanding Academic Performance (12/2022) First Prize Scholarship of Academic science and technology (12/2021) • Courses: Machine Learning, Pattern Recognition, Big Data Analysis, Numerical Analysis, Natural Language Processing.	Sep 2019 - Jun 2023

WORK EXPERIENCE

Fondazione Bruno Kessler - FBK <i>Research Intern</i>	Mar 2025 - Mar 2026 Trentino-Alto Adige, Italy
- Proposed a new embodied perception task and built the first offline multi-view benchmark for active instance verification in realistic 3D environments. - Designed an MLLM-based agent with attribute decomposition, confidence-weighted multi-view tracking, and next-best-view planning; fine-tuned via LoRA with SFT→RL alignment, achieving 85.6% accuracy.	
Chengdu Jiaoda Guangmang Technology Co., Ltd. <i>Algorithm Intern</i>	Jul 2022 - Sep 2022 Chengdu, Sichuan, China
- Researched the newest machine learning algorithms and industrial anomaly detection algorithm. - Applied GAN models to reconstruct images to identify anomalies and evaluate the performance. - Used the object detection neural network model to detect abnormal parts in high-speed rail supports. - Assisted colleagues in developing a set of character recognition patents for LCD meters.	

COMPETITIONS

Kaggle March Machine Learning Mania 2025 - Silver Medal (64th / 1,727) <i>Solo</i>	Apr 2025
Kaggle BirdCLEF 2024 Competition - Bronze Medal (70th / 974) <i>Solo</i>	Jun 2024
Kaggle BirdCLEF+ 2025 Competition - Bronze Medal (199th / 2,025) <i>Solo</i>	Jun 2025
Mathematical Contest In Modeling (MCM), Finalist winner (top 1%) <i>Leader</i>	Apr 2021
- Designed a set of algorithms to assess the degree of hunger around the world and optimize the food supply chain. - Responsible for modeling, programming and part of paper writing. - The youngest winner in the history of Anhui University and the first winner of School of Internet.	
iCAN Innovation Contest 2021 (Anhui) - Second prize <i>Team member</i>	Oct 2021
- Developed a set of hardware system capable of intelligently monitoring and warning the abnormal behavior of the elderly. - Responsible for some STM32 development and project plan writing.	

PUBLICATIONS

"CL-RAG: A Closed-Loop Multimodal Retrieval-Augmented Generation Architecture for Robust Human-Robot Control Interaction"

WRC SARA 2025 (oral)

Bowen Zhang, **Yuhang Jiang**, Lingxiang Hu, Dun Li, Qianqian Hu*

"Improved lightweight identification of agricultural diseases based on MobileNetV3"

CAIBDA 2022 (oral)

Yuhang Jiang*, Wenping Tong

Software Copyright of Intelligent contract software for agricultural insurance compensation

2022SR1568111.

Note: * indicates the corresponding author.

RESEARCH EXPERIENCE

KDEN-NSGA-II Scheduler for Flexible Job-Shop Scheduling

Jun 2025

- Developed a multi-objective FJSP solver based on enhanced NSGA-II with hybrid initialization and adaptive crossover, optimizing makespan, load balance, and machine idle time.
- Built a modular benchmarking framework supporting multiple baselines (NSGA-II, NSGA-III, SPEA2, multi-objective ACO) with KNN-based density selection for diversity-aware search.

High Performance Computing for Grey Wolf Optimizer (GWO) Optimization

Jan 2025

- Proposed HGT-GWO, integrating global historical best positions and trend guidance into GWO, outperforming standard GWO on 15 benchmark functions.
- Implemented a master-worker island parallelization scheme in C/MPI/OpenMP on UNITN's HPC cluster, reducing communication overhead through controlled synchronization.

Augmented Reality-Driven Robotic Arm Control for Industrial Automation

Sep 2024

- Built an AR-based robotic arm control system on Microsoft HoloLens 2, combining YOLOv8 and FastSAM for real-time gesture recognition and fingertip coordinate mapping.
- Supported multi-mode operation (gesture-based and interface-based) with robustness to hand occlusion in industrial environments.

Research on Semantic Segmentation Method of High-Resolution Remote Sensing Images

May 2023

Outstanding Undergraduate Graduation Project

- Developed an encoder-decoder model with residual-weighted attention for remote sensing segmentation, achieving F1-scores of 90.23% and 87.37% on ISPRS Potsdam and Vaihingen datasets.

SKILLS LIST

- **Programming:** Python, C/C++, Matlab, R, Java, C#.
- **Toolkits:** Pytorch, Tensorflow, Paddle Paddle, Sklearn, MPI, OpenMP etc.
- **Deep Learning Architectures:** CNN, RNN, Transformer, Mamba.
- **Algorithms:** Computer Vision, Natural Language Processing, ARIMA, Reinforcement Learning, Evolutionary Algorithms.
- **Applications:** Image Classification, Image Segmentation, Object Tracking, Industrial Anomaly Detection, Time Series Data Analysis.
- **Other Skills:** Unity, Quality Management, Lean Manufacturing, Circular Economy, Sustainable Management, etc.

LANGUAGE

- **English:** Fluent, TOEFL 102
- **Chinese:** Native Speaker